Metro Systems Corporation
Public Company Limited

Course:

Advanced DevSecOps Pipeline Implementation & Management

Duration: 2 Days

Course Overview

This "Advanced DevSecOps Pipeline Implementation & Management" course is designed to equip participants with a deep understanding and practical skills in building a comprehensive DevSecOps pipeline. Learners will cover everything from secure source code management, CI/CD, and vulnerability scanning, to secure application deployment using industry-leading tools like GitLab, Jenkins, SonarQube, Harbor, Argo CD, and HashiCorp Vault. By the end of the course, participants will gain the expertise needed to integrate security into every phase of the software development lifecycle, ensuring the delivery of high-quality and secure applications to production.



Course Structure (Modules):

Module 0: Introduction to DevSecOps & Pipeline Overview

- DevOps vs. DevSecOps: Why DevSecOps?
- The Importance of Security at Every Stage of the SDLC.
- Overview of the DevSecOps Pipeline based on the provided Diagram (What each tool does).
- Advantages and Benefits of a Complete DevSecOps Pipeline.
- Overview of the Infrastructure and Environment used for the training (e.g., using VM or Cloud Lab Environment).

Module 1: Source Code Management & Static Analysis Security Testing (SAST)

GitLab:

- Using GitLab for Version Control (Repo, Branching, Merge Requests).
- Setting up Webhooks for Integration.
- Hands-on Lab: Create a GitLab Repository, Push Code.

SonarQube:

- SAST Concepts and SonarQube's functionality.
- Installation and configuration of SonarQube Server and SonarScanner.
- Integrating SonarQube with GitLab (Webhooks/Integration).
- Hands-on Lab: Install SonarQube, set up a project in SonarQube, scan code, and analyze results.

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Course Structure (Modules):

Module 2: Continuous Integration (CI) with Jenkins & Containerization

Jenkins:

- CI Principles and the Role of Jenkins.
- Installation and configuration of Jenkins Master/Agent.
- Creating Jenkins Pipelines (Jenkinsfile) for Build and Test.
- Integrating Jenkins with GitLab (SCM polling, Webhooks).
- Hands-on Lab: Install Jenkins, create a Jenkinsfile for project build, connect to GitLab.

Harbor Container Registry:

- The Importance of a Secure Container Registry.
- Installation and configuration of Harbor.
- Pushing/Pulling Docker Images from Jenkins to Harbor.
- Hands-on Lab: Install Harbor, configure Jenkins to push images to Harbor.

Module 3: Automated Testing & Secret Management

Selenium:

- Automated UI Testing with Selenium.
- Integrating Selenium Tests into the Jenkins Pipeline.
- Hands-on Lab: Write simple test cases with Selenium, run tests via Jenkins.

Apache JMeter:

- Load Testing Concepts and JMeter Usage.
- Creating Test Plans and Scenarios.
- Integrating JMeter Tests into the Jenkins Pipeline.
- Hands-on Lab: Create a simple load test with JMeter, run tests via Jenkins.

HashiCorp Vault:

- The Importance of Secret Management in DevSecOps.
- Installation and configuration of HashiCorp Vault.
- Storing and retrieving Secrets from Vault in Jenkins/Argo CD.
- Hands-on Lab: Install Vault, store and retrieve secrets in a Jenkins Pipeline.

Module 4: Continuous Delivery (CD) with Argo CD & Deployment

Argo CD:

- GitOps Principles and the Role of Argo CD.
- Installation and configuration of Argo CD on a Kubernetes Cluster.
- Creating an Argo CD Application for Deployment from Harbor/GitLab.
- Integrating Argo CD with Jenkins (Trigger CD).
- Hands-on Lab: Install Argo CD, configure Argo CD to sync with a GitLab Repo (containing Kubernetes Manifests), deploy an application.

Deployment to Cloud Platforms:

- Concepts of deploying to Kubernetes, Azure, AWS.
- Using Kubernetes Manifests (YAML).
- Hands-on Lab: Deploy a sample application to MiniKube/kind or a designated Cloud Provider (e.g., AKS/EKS if environments are available).